



Curriculum Vitae

Salah Faisal Saeed Saeed

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Sex Male | Date of birth 12/11/1998 | Nationality Yemeni

PERSONAL STATEMENT

Mechatronics and Robotics Engineer, STEM/Physics educator, and researcher with experience in technical education, laboratory development, electronics, embedded systems, robotics, industrial control, computer systems, and engineering simulation. I hold a B.Sc. in Mechatronics and Robotics Engineering and a Vocational Professional Diploma in Industrial Control Systems / Electronics, completed with 92.27% (Excellent), ranking first in my cohort and second nationally in Yemen. My background combines secondary-level Physics teaching with hands-on project work in Arduino, ESP32, sensors, IoT, C++, circuit simulation, PCB design, and MATLAB/Simulink, along with computer and laboratory maintenance and industrial exposure to PLC-based systems. I also contribute to research in robotics, computer vision, EV charging, artificial intelligence, and related engineering fields.

WORK EXPERIENCE

01/2022 – Present

STEM Technical Education Specialist & Laboratory Engineer (Full-time - Onsite) **Al-Shaheeda Ne'mah Ahmed Rassam Basic and Secondary School for Girls, Taiz, Yemen**

- Delivered practical training for physics teachers in electronics, measurement, circuit construction, Multisim, and related applications.
- Established and developed the school's Robotics and Prototyping Laboratory and supported its technical setup and activities.
- Trained students in Arduino, ESP32, embedded systems, robotics, IoT, C++, and PCB design using Proteus.
- Trained students in MATLAB/Simulink and Artificial Bee Colony optimization for engineering simulation and projects.
- Guided students in technical projects, competitions, research, testing, presentations, and academic writing.

Teaching and Training Areas:

- Electronics, Electrical Fundamentals, and Measurement
- Analog and Digital Circuits
- Circuit Construction and Multisim Simulation
- Arduino, ESP32, Sensors, and Motors
- Embedded Systems, Robotics, and IoT
- C++ Programming Fundamentals
- PCB Design using Proteus
- MATLAB/Simulink Modeling and Simulation
- Artificial Bee Colony Optimization with MATLAB
- Robotics and Prototyping Projects
- Scientific Research and Academic Writing

08/2023 – 05/2025

Physics Teacher – Second Secondary (Grade 11) (Part-time - Onsite) **Al-Shaheeda Ne'mah Ahmed Rassam Basic and Secondary School for Girls, Taiz, Yemen**

- Taught Grade 11 Physics and supported laboratory experiments, demonstrations, and hands-on activities.
- Connected theoretical concepts with practical observation, measurement, and experimental work.
- Supported the safe and effective use of physics laboratory equipment and resources.

2023–2024

Computer Maintenance Engineer & Physics Laboratory Technician (Part-time) **AL-ASBAHI Private School, Taiz, Yemen**

- Performed preventive and corrective maintenance of computer systems, operating environments, educational software, and basic network connectivity.
- Prepared physics laboratory experiments, organized equipment, supported practical sessions, and maintained safe experimental procedures.

- Provided technical support to staff and students while improving the reliability of classroom and laboratory infrastructure.

2022–2023 Physics Teacher & Laboratory Support Instructor (Part-time)
Al-Ghad Al-Mushreq School, Taiz, Yemen

- Taught physics to secondary-level students, including first, second, and third secondary grades.
- Supported physics laboratory activities through experiment preparation, practical demonstrations, and safe laboratory practice.
- Contributed to the preparation and development of physics laboratory equipment and electronic experimental setups.
- Performed basic maintenance and troubleshooting of laboratory devices to support continuity of practical learning.

2018–2019 Industrial Training Assistant — Industrial Control Systems(Part-time)
Technical Industrial Institute, Taiz, Yemen

- Assisted in practical training activities related to industrial control systems and electronics.
- Supported hands-on instruction in control circuits, electronic components, panel-level practice, and troubleshooting.
- Contributed to vocational laboratory sessions and strengthened early exposure to applied industrial-control training.

EDUCATION

07/2025 B.Sc. in Engineering (Mechatronics and Robotics Engineering)
Taiz University, Alsaeed Faculty for Engineering & Information Technology, Taiz, Yemen

Cumulative grade: Good

Medium of Instruction: English (official university confirmation letter available).

Graduation project: Manipulator Robot Error Correction Using Computer Vision

2016/2017 Vocational Professional Diploma (Industrial Control Systems / Electronics)
Technical Industrial Institute (Baghad.st), Sana'a, Yemen

Cumulative grade: 92.27% (Excellent). Ranked Second Republic.

JOB-RELATED SKILLS

- **STEM & Physics Education:** Secondary-level Physics teaching, STEM/STEAM instruction, laboratory-based learning, technical training, and student project supervision.
- **Robotics & Embedded Systems:** Arduino, ESP32, microcontrollers, sensors, actuators, IoT, C++ fundamentals, and robotics prototyping.
- **Electronics & Industrial Control:** Analog and digital electronics, electrical measurements, circuit construction, industrial control fundamentals, and PLC-based systems.
- **Engineering Software:** MATLAB/Simulink, Multisim, Proteus, circuit simulation, PCB design, and engineering modeling.
- **Technical & Laboratory Support:** Laboratory setup, equipment organization, computer maintenance, troubleshooting, and basic technical maintenance.
- **Research & Technical Writing:** Engineering research, literature review, academic writing, technical reports, presentations, and research project development.
- **Computer & Digital Skills:** Computer hardware/software support, basic networking, Microsoft Office, and educational technology.

PERSONAL SKILLS

- Strong communication and interpersonal skills
- Teamwork and collaboration
- Problem-solving and analytical thinking
- Organization and time management
- Adaptability and willingness to learn
- Responsibility and attention to detail

- Leadership and student mentoring
- Ability to work independently and under pressure

LANGUAGES

Arabic (Native)

English (Intermediate)

TRAINING & COURSES

- 05/2026 **IEEE Authorship and Open Access Symposium: Tips and Best Practices to Get Published from IEEE Editors**
IEEE — Certificate of Participation, 90 minutes.
- 09/2021 **Training Course in Translation – Al-Kindi Institute for Languages & Computer, Yemen-Taiz**
60 Hours (Grade: Excellent – 98%)(issued 25/02/2021)
- 05/2026 **Train-the-Trainer Program: The Novice Trainer Level**
Edraak — International Board of Certified Trainers (IBCT) — Level Completion Certificate
- 02/2024 **Internet of Things (IoT) — Edraak Platform**
Level Completion Certificate (issued 25/02/2024).
- 02/2024 **Influential Leadership & Effective Management— Edraak Platform**
Level Completion Certificate (issued 24/02/2024).

Industrial & Technical Training

- 08/2024 **Cooperative Industrial Training — General Industries & Packages Co. (GenPack), Taiz, Yemen**
Gained practical exposure to packaging production workflows, industrial operations, workplace discipline, and safety practices.
- 08/2024–09/2024 **Industrial Training — National Dairy & Food Co. (Nadfood), Taiz, Yemen**
Completed training in the Engineering Department with exposure to plant utilities, production systems, maintenance operations, PLC-based control, and major industrial equipment.
- 01/2018 **Electronic Control Training — Yemen Company for Industry & Commerce (YCIC – HSA Group), Yemen**
Completed practical training in electronic control systems, industrial control applications, and panel-level technical work.

AWARDS & RECOGNITION

- 09/2026 **Certificate of Appreciation — Excellence in Scientific Research**
Taiz University, Taiz, Yemen
Awarded in recognition of outstanding contributions to scientific research and scholarly work, and for supporting the University's academic presence within the international research community.
- 08/2026 **Best Paper Award — eSmarTA-2026**
6th International Conference on Emerging Smart Technologies and Applications, Thamar University, Yemen.
Received for the paper "Toward Trust-by-Design in Federated Learning for Healthcare: A Taxonomic Review of Privacy, Explainability, and Accountability."
- 08/2026 **eSmarTA-2026 Conference Participation and Presentations**
6th International Conference on Emerging Smart Technologies and Applications, Thamar University, Yemen

Participated online in eSmarTA-2026 and co-authored four papers that were presented at the conference, covering EV fast-charging and power electronics, federated learning in healthcare, and deep learning for computational imaging.

08/2025 eSmarTA-2025 Conference Participation and Presentations

5th International Conference on Emerging Smart Technologies and Applications, Ibb University, Yemen

Participated online in eSmarTA-2025 and co-authored four papers that were presented at the conference, covering camera-based measurement systems, robotic-arm technologies, robotic-manipulator error compensation, and transcranial direct current stimulation.

2016/2017 Academic Distinction — Ranked First in Cohort and Second Nationally

Technical Industrial Institute, Sana'a, Yemen

Awarded for outstanding performance in Industrial Control Systems / Electronics with a cumulative grade of 92.27% (Excellent).

2014–2016 Academic Distinction — First Rank in Cohort

Technical Industrial Institute, Al-Hasab, Taiz, Yemen.

Recognized for excellent academic and practical performance during the first and second vocational levels in Industrial Control Systems / Electronics.

2026 Certificates of Appreciation — Laboratory Development, Robotics & Prototyping, Research Mentorship, and Professional Service

Al-Shaheeda Ne'mah Ahmed Rassam Basic and Secondary School for Girls, Taiz, Yemen

Recognized for contributions to laboratory development, the establishment of the Robotics and Prototyping Laboratory, student research mentorship, technical training, STEM education, and hands-on project support. The recognition also covered laboratory setup, equipment organization, troubleshooting, maintenance, prototyping, testing, and supporting students in practical engineering and research activities.

2023 Certificates of Appreciation — Teaching Excellence and Physics Laboratory Development

Al-Ghad Al-Mushreq School, Taiz, Yemen

Received for contributions to secondary-level physics teaching, student academic guidance, physics laboratory equipment preparation, electronic experiments, and hands-on practical learning.

PUBLICATIONS

1. **S. F. S. Saeed**, R. A. A. Saleh, A. Haidar, M. Ghaleb, H. AbuAlkebash and M. A. A. S "Control-Oriented Design and Performance Evaluation of a Three-Terminal Storage-Assisted EV Fast-Charging System with an Interleaved Bidirectional Buck–Boost Converter," 2026 6th International Conference on Emerging Smart Technologies and Applications (eSmarTA), Dhamar, Yemen, 2026, pp. 1-10, doi: <https://doi.org/10.1109/eSmarTA70636.2026.11652324>
2. R. A. A. Saleh, **S. F. S. Saeed**, M. A. H. Al-Ariqi, M. E. T. Ahmed, M. M. A. Yahya and Ghaleb, "Toward Trust-by-Design in Federated Learning for Healthcare: A Taxonomic Review of Privacy, Explainability, and Accountability," 2026 6th International Conference on Emerging Smart Technologies and Applications (eSmarTA), Dhamar, Yemen, 2026, pp. 1-10, doi: <https://doi.org/10.1109/eSmarTA70636.2026.11652200>
3. M. A. A. Saif, **S. F. S. Saeed**, A. Haidar, A. F. M. Almakhathay, M. M. N. Al-Koshab and H. A. Qasem, "A Review of Electric Vehicle Charging Systems: Converter Architecture Control Optimization, and PV-Integrated Energy Management," 2026 6th International Conference on Emerging Smart Technologies and Applications (eSmarTA), Dhamar, Yemen, 2026, pp. 1-10, doi: <https://doi.org/10.1109/eSmarTA70636.2026.11652317>
4. Radhwan A. A. **Saleh; Salah F. S. Saeed**; Malak M. N. Al-Koshab; Alzahraa F. M. Almakhathay; Abdullah Haidar; Kerem Küçük, "Data-Driven Deep Learning for Imaging

Through Scattering Media: Architectures, Learning Paradigms, and Methodological Frontiers," 2026 6th International Conference on Emerging Smart Technologies and Applications (eSmarTA), Dhamar, Yemen, 2026, pp. 1-10, doi: <https://doi.org/10.1109/eSmarTA70636.2026.11652149>

5. Saleh, R. A., **Nasser, S. F.**, & Abdullah, T. S. (2025, August). Advances in Error Compensation for Robotic Manipulators: A Systematic Review from Geometric Calibration to AI-Driven Hybrid Control. In *2025 5th International Conference on Emerging Smart Technologies and Applications (eSmarTA)* (pp. 1-10). IEEE. <https://doi.org/10.1109/eSmarTA66764.2025.11131700>
6. Saleh, R. A., **Nasser, S. F.**, Al Mashwali, M. M., Taha, E. M., & Al-Noor, O. A. (2025, August). A Survey on Camera-Based Measurement Systems: Techniques, Application and Challenges. In *2025 5th International Conference on Emerging Smart Technologies and Applications (eSmarTA)* (pp. 1-9). IEEE. <https://doi.org/10.1109/eSmarTA66764.2025.11132174>
7. Saleh, R. A., **Nasser, S. F.**, Almakthay, A. F., & Al-Koshab, M. M. (2025, August). A Comprehensive Review of Transcranial Direct Current Stimulation (tDCS): Mechanism, Cognitive Effects, Applications, and Future Directions. In *2025 5th International Conference on Emerging Smart Technologies and Applications (eSmarTA)* (pp. 1-8). IEEE. <https://doi.org/10.1109/eSmarTA66764.2025.11132143>
8. Saleh, R. A., Alqudsi, Y., Al-Hajj, A. A., Abdullah, A. A., **Nasser, S. F.**, & Amrani, M. (2025, August). Advancements in Robotic Arm Technologies: Precision, Intelligence, Cross-Sector Applications. In *2025 5th International Conference on Emerging Smart Technologies and Applications (eSmarTA)* (pp. 1-10). IEEE. <https://doi.org/10.1109/eSmarTA66764.2025.11132178>

UNDER WRITING

1. Hybrid Geometric–Sliding Mode Control of Quadrotor UAVs with Disturbance Rejection Optimized.
2. Optimization-Assisted Model Predictive Control for Grid-Compliant PV-Integrated Bidirectional EV Fast-Charging Systems.
3. Reliability-Constrained Multi-Objective Optimization of an Off-Grid PV–Wind–Diesel–Battery Hybrid Energy System: A HOMER-Benchmarked Cost–Renewability Assessment.
4. Explainable and Privacy-Preserving Federated Learning in Healthcare: A Systematic Review and Taxonomic Synthesis.
5. Security Digital Twins for SOC Engineering: Threat Simulation, Detection Content Generation, and Response Validation: A Systematic Review.
6. Automating Cybersecurity Penetration Testing Tasks with AI Agents Using LLMs: A Systematic Review.

RESEARCH OF INTERESTS

1. Robotics, computer vision, and intelligent control systems.
2. MATLAB/Simulink-based modeling, simulation, and optimization.
3. EV fast-charging systems and renewable-energy-integrated power conversion.
4. Federated learning, explainable AI, and trustworthy AI.
5. Cybersecurity, IoT security, and security digital twins.
6. Metaheuristic optimization algorithms and engineering applications.

REFERENCES

1. Researches and graduation project supervisor Ass. Prof. Dr. Radhwan Ali Abdulghani Saleh, Assistant Professor, Department of Software Engineering, Kocaeli University, Türkiye: radhwan.saleh@kocaeli.edu.tr
2. Research co-supervisor Dr. Mubaarak A. A. Saif, Department of Energy Systems Engineering, Faculty of Technology, Kocaeli University, Türkiye: mubarak.saif@kocaeli.edu.tr
3. Research co-supervisor Ass. Prof. Dr. Mustafa Ghaleb, Assistant Professor, Department of Software Engineering, Faculty of Engineering, Kocaeli University, Türkiye: Mustafa.ghaleb@kocaeli.edu.tr
4. Professional reference T. Rajaa Al-Dubai, Manager, Al-Shaheeda Na'mah Rassam School, Taiz, Yemen: ne3marasam@ne3marasam.org